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CODE:

```
# Future Sales Prediction using Linear Regression

import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Sample sales dataset
data = {
    "Month": [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12],
    "Sales": [
        12000, 15000, 18000, 21000,
        24000, 27000, 30000, 32000,
        35000, 38000, 41000, 45000
    ]
}

# Convert data into DataFrame
df = pd.DataFrame(data)

print("SALES DATASET")
print("=" * 40)
print(df)

# Input feature
X = df[["Month"]]

# Target variable
y = df["Sales"]

# Create Linear Regression model
model = LinearRegression()

# Train the model
model.fit(X, y)

# Predict sales for existing months
y_pred = model.predict(X)

# Evaluate the model
mse = mean_squared_error(y, y_pred)
r2 = r2_score(y, y_pred)

print("\nMODEL PERFORMANCE")
print("=" * 40)
print("Mean Squared Error:", round(mse, 2))
print("R-squared Score:", round(r2, 4))

# Predict future sales
future_months = pd.DataFrame({
    "Month": [13, 14, 15, 16]
})

future_sales = model.predict(future_months)

print("\nFUTURE SALES PREDICTION")
print("=" * 40)
```

```python
for month, sales in zip(future_months["Month"], future_sales):
    print(
        "Month:",
        month,
        "Predicted Sales: ₹",
        round(sales, 2)
    )
```

OUTPUT :

```
SALES DATASET
========================================
    Month  Sales
0       1  12000
1       2  15000
2       3  18000
3       4  21000
4       5  24000
5       6  27000
6       7  30000
7       8  32000
8       9  35000
9      10  38000
10     11  41000
11     12  45000

MODEL PERFORMANCE
========================================
Mean Squared Error: 138306.14
R-squared Score: 0.9986

FUTURE SALES PREDICTION
========================================
Month: 13 Predicted Sales: ₹ 47121.21
Month: 14 Predicted Sales: ₹ 50037.3
Month: 15 Predicted Sales: ₹ 52953.38
Month: 16 Predicted Sales: ₹ 55869.46
```